ELEVATE LABS
PROJECT -1
Linux Hardening Audit Tool

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Domain: Cybersecurity

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Objective: Create a tool to audit a Linux system's security configuration.

1. **Check firewall rules, unused services, SSH settings.**
   A firewall is a network security mechanism that controls incoming and outgoing traffic based on predefined security rules. Proper firewall configuration helps prevent unauthorized access while allowing legitimate traffic. Reviewing firewall rules ensures that only required ports and services are open. Unnecessary open ports increase the attack surface and make the system vulnerable to attacks such as port scanning and exploitation. Regularly checking firewall rules helps maintain system security and compliance with security policies.

   Check firewall status (UFW – Ubuntu/Kali)
   sudo ufw status verbose

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ sudo ufw status verbose

[sudo] password for kali:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22                         ALLOW IN    Anywhere
80                         ALLOW IN    Anywhere
443                        ALLOW IN    Anywhere
Anywhere                   DENY IN     192.168.10.3
22                         ALLOW IN    192.168.1.10
80/tcp                     ALLOW IN    Anywhere
23                         DENY IN     Anywhere
21/tcp                     DENY IN     Anywhere
Anywhere                   DENY IN     192.168.1.50
Anywhere                   DENY IN     192.168.1.100
22/tcp                     ALLOW IN    Anywhere
22 (v6)                    ALLOW IN    Anywhere (v6)
80 (v6)                    ALLOW IN    Anywhere (v6)
443 (v6)                   ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
23 (v6)                    DENY IN     Anywhere (v6)
21/tcp (v6)                DENY IN     Anywhere (v6)
22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

List all UFW rules

sudo ufw status numbered

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw status numbered

Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22                         ALLOW IN    Anywhere
[ 2] 80                         ALLOW IN    Anywhere
[ 3] 443                        ALLOW IN    Anywhere
[ 4] Anywhere                   DENY IN     192.168.10.3
[ 5] 22                         ALLOW IN    192.168.1.10
[ 6] 80/tcp                     ALLOW IN    Anywhere
[ 7] 23                         DENY IN     Anywhere
[ 8] 21/tcp                     DENY IN     Anywhere
[ 9] Anywhere                   DENY IN     192.168.1.50
[10] Anywhere                   DENY IN     192.168.1.100
[11] 22/tcp                     ALLOW IN    Anywhere
[12] 22 (v6)                    ALLOW IN    Anywhere (v6)
[13] 80 (v6)                    ALLOW IN    Anywhere (v6)
[14] 443 (v6)                   ALLOW IN    Anywhere (v6)
[15] 80/tcp (v6)                ALLOW IN    Anywhere (v6)
[16] 23 (v6)                    DENY IN     Anywhere (v6)
[17] 21/tcp (v6)                DENY IN     Anywhere (v6)
[18] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

Check iptables rules (CentOS/older systems)

sudo iptables -L -v -n

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L -v -n

Chain INPUT (policy DROP 2 packets, 1270 bytes)
 pkts bytes target     prot opt in     out     source               destination
    9  3257 ufw-before-logging-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    9  3257 ufw-before-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    2  1270 ufw-after-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    2  1270 ufw-after-logging-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    2  1270 ufw-reject-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    2  1270 ufw-track-input  all  --  *      *       0.0.0.0/0            0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DOCKER-USER  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 DOCKER-ISOLATION-STAGE-1  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ACCEPT     all  --  *      docker0  0.0.0.0/0            0.0.0.0/0            ctstate RELATED,ESTABLISHED
    0     0 DOCKER     all  --  *      docker0  0.0.0.0/0            0.0.0.0/0
    0     0 ACCEPT     all  --  docker0 !docker0  0.0.0.0/0            0.0.0.0/0
    0     0 ACCEPT     all  --  docker0 docker0  0.0.0.0/0            0.0.0.0/0
    0     0 ufw-before-logging-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ufw-before-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ufw-after-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ufw-after-logging-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ufw-reject-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
    0     0 ufw-track-forward  all  --  *      *       0.0.0.0/0            0.0.0.0/0
```

Check firewalld rules (RHEL/CentOS)

sudo firewall-cmd --list-all

```
┌──(kali㉿kali)-[~]
└─$ sudo firewall-cmd --list-all

public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: eth0 eth1
  sources:
  services: dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Allow only SSH (example)

sudo ufw allow 22/tcp

sudo ufw enable

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw allow 22/tcp
sudo ufw enable

Skipping adding existing rule
Skipping adding existing rule (v6)
Firewall is active and enabled on system startup
```

Check Unused Services

Unused or unnecessary services running on a system can create security vulnerabilities. Attackers often exploit outdated or misconfigured services to gain unauthorized access. Minimizing running services reduces the system's attack surface and improves performance. It is important to identify active services, evaluate whether they are required, and disable those that are not needed.

List running services

sudo systemctl list-units --type=service --state=running

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl list-units --type=service --state=running

  UNIT                          LOAD   ACTIVE SUB     DESCRIPTION
  accounts-daemon.service       loaded active running Accounts Service
  apache2.service               loaded active running The Apache HTTP Server
  colord.service                loaded active running Manage, Install and Generate Color Profiles
  containerd.service            loaded active running containerd container runtime
  cron.service                  loaded active running Regular background program processing daemon
  dbus.service                  loaded active running D-Bus System Message Bus
  docker.service                loaded active running Docker Application Container Engine
  fail2ban.service              loaded active running Fail2Ban Service
  firewalld.service             loaded active running firewalld - dynamic firewall daemon
  getty@tty1.service            loaded active running Getty on tty1
  haveged.service               loaded active running Entropy Daemon based on the HAVEGE algorithm
  lightdm.service               loaded active running Light Display Manager
  mariadb.service               loaded active running MariaDB 11.8.5 database server
  ModemManager.service          loaded active running Modem Manager
  NetworkManager.service        loaded active running Network Manager
  polkit.service                loaded active running Authorization Manager
  rtkit-daemon.service          loaded active running RealtimeKit Scheduling Policy Service
  ssh.service                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service      loaded active running Journal Service
  systemd-logind.service        loaded active running User Login Management
  systemd-udevd.service         loaded active running Rule-based Manager for Device Events and Files
  systemd-userdbd.service       loaded active running User Database Manager
  udisks2.service               loaded active running Disk Manager
  unattended-upgrades.service   loaded active running Unattended Upgrades Shutdown
  upower.service                loaded active running Daemon for power management
  user@1000.service             loaded active running User Manager for UID 1000
```

Check all enabled services

sudo systemctl list-unit-files --type=service | grep enabled

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl list-unit-files --type=service | grep enabled

accounts-daemon.service                    enabled         enabled
apache2.service                            enabled         disabled
console-setup.service                      enabled         enabled
cron.service                               enabled         enabled
docker.service                             enabled         enabled
fail2ban.service                           enabled         disabled
firewalld.service                          enabled         disabled
getty@.service                             enabled         enabled
grub-install-devices.service               enabled         disabled
haveged.service                            enabled         enabled
keyboard-setup.service                     enabled         enabled
lightdm.service                            enabled         disabled
mariadb.service                            enabled         disabled
ModemManager.service                       enabled         enabled
```

Stop a service

sudo systemctl stop apache2

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl stop apache2
```

Disable a service permanently

sudo systemctl disable apache2

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl disable apache2

Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install
Executing: /usr/lib/systemd/systemd-sysv-install disable apache2
Removed '/etc/systemd/system/multi-user.target.wants/apache2.service'.
```

Check listening ports

sudo ss -tulnp

```
┌──(kali㉿kali)-[~]
└─$ sudo ss -tulnp

Netid       State       Recv-Q      Send-Q      Local Address:Port      Peer Address:Port      Process
tcp         LISTEN      0           80              127.0.0.1:3306          0.0.0.0:*          users:(("mariadbd",pid=857,fd=27))
tcp         LISTEN      0           128               0.0.0.0:22            0.0.0.0:*          users:(("sshd",pid=779,fd=6))
tcp         LISTEN      0           4096            127.0.0.1:35403         0.0.0.0:*          users:(("containerd",pid=772,fd=9))
tcp         LISTEN      0           128                  [::]:22             [::]:*          users:(("sshd",pid=779,fd=7))

┌──(kali㉿kali)-[~]
```

Check SSH Settings

Secure Shell (SSH) allows remote access to a system. Improper SSH configuration can lead to brute-force attacks and unauthorized access. Best practices include disabling root login, changing the default SSH port, disabling password authentication in favor of key-based authentication, and limiting access to specific users. Reviewing SSH settings ensures secure remote administration and reduces the risk of compromise.

Check SSH configuration file

sudo nano /etc/ssh/sshd_config

Important settings to verify:

PermitRootLogin no

PasswordAuthentication no

Port 22

AllowUsers username

```
  GNU nano 8.7                                                    /etc/ssh/sshd_config
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication.  Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no
```

Restart SSH service after changes

sudo systemctl restart ssh

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart ssh
```

Check SSH service status

sudo systemctl status ssh

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status ssh

● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-02-19 08:40:41 EST; 29s ago
 Invocation: 43a959b33b7b4dbe9d14d634dca3ed58
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 9313 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 9315 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 1.5M (peak: 2.1M)
        CPU: 26ms
     CGroup: /system.slice/ssh.service
             └─9315 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 19 08:40:41 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 19 08:40:41 kali sshd[9315]: Server listening on 0.0.0.0 port 22.
Feb 19 08:40:41 kali sshd[9315]: Server listening on :: port 22.
Feb 19 08:40:41 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Check Failed SSH Logins using journalctl

sudo journalctl -u ssh

```
┌──(kali㉿kali)-[~]
└─$ sudo journalctl -u ssh

Feb 03 10:45:42 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
Feb 03 10:46:46 kali sshd-session[3835]: Connection closed by 127.0.0.1 port 49538 [preauth]
Feb 03 10:47:43 kali sshd-session[4072]: Connection closed by 127.0.0.1 port 33820 [preauth]
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM err
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'ka
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

2. Verify permissions on key files (/etc/shadow, /etc/passwd).

The files /etc/passwd and /etc/shadow are critical system files in Linux that store user account information. The /etc/passwd file contains basic user details such as username, UID, GID, home directory, and shell. It must be readable by all users because many system processes require access to this information. However, it should only be writable by the root user to prevent unauthorized modifications.

The /etc/shadow file stores encrypted password hashes and password aging information. Because it contains sensitive authentication data, it must be accessible only to the root user. Incorrect permissions on these files can allow attackers to modify user accounts or extract password hashes, leading to privilege escalation or system compromise. Therefore, verifying and maintaining correct permissions is an essential system

Check Permissions of /etc/passwd
ls -l /etc/passwd

```
┌──(kali㉿kali)-[~]
└─$ ls -l /etc/passwd

-rw-r--r-- 1 root root 3433 Feb 15 06:21 /etc/passwd
```

Check Permissions of /etc/shadow
sudo ls -l /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ sudo ls -l /etc/shadow

-rw-r----- 1 root shadow 1510 Feb 15 06:21 /etc/shadow
```

Verify Ownership
ls -l /etc/passwd /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ ls -l /etc/passwd /etc/shadow

-rw-r--r-- 1 root root   3433 Feb 15 06:21 /etc/passwd
-rw-r----- 1 root shadow 1510 Feb 15 06:21 /etc/shadow
```

Fix Incorrect Permissions (If Needed)
Set correct permissions:
sudo chmod 644 /etc/passwd
sudo chmod 640 /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ sudo chmod 644 /etc/passwd
sudo chmod 640 /etc/shadow
```

Set correct ownership:
sudo chown root:root /etc/passwd
sudo chown root:shadow /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ sudo chown root:root /etc/passwd
sudo chown root:shadow /etc/shadow
```

Check for Unauthorized Changes
sudo stat /etc/passwd
sudo stat /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ sudo stat /etc/passwd
sudo stat /etc/shadow

  File: /etc/passwd
  Size: 3433            Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4984507     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-02-19 08:49:48.444848142 -0500
Modify: 2026-02-15 06:21:58.618940455 -0500
Change: 2026-02-19 08:49:48.440846142 -0500
 Birth: 2026-02-15 06:21:58.616374385 -0500
  File: /etc/shadow
  Size: 1510            Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4981083     Links: 1
Access: (0640/-rw-r-----)  Uid: (    0/    root)   Gid: (   42/  shadow)
Access: 2026-02-19 08:50:22.367811419 -0500
Modify: 2026-02-15 06:21:58.626207316 -0500
Change: 2026-02-19 08:49:48.448850142 -0500
 Birth: 2026-02-15 06:21:58.624085561 -0500
```

3. **Check for rootkit indicators**
A rootkit is a type of malicious software designed to gain unauthorized root or administrative access to a system while hiding its presence. Rootkits modify system files, processes, kernel modules, or logs to avoid detection. They may allow attackers to maintain persistent access, steal data, or control the system remotely. Detecting rootkits is difficult because they attempt to conceal their activity. Therefore, system administrators must regularly check for rootkit indicators such as hidden processes, suspicious kernel modules, unexpected open ports, modified system binaries, and integrity violations. Tools like rkhunter and chkrootkit are commonly used in Linux systems to scan for known rootkit signatures and suspicious behavior.

4. Install Rootkit Detection Tools (Kali/Ubuntu)

Install rkhunter

sudo apt update

sudo apt install rkhunter -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
sudo apt install rkhunter -y

Hit:1 http://http.kali.org/kali kali-rolling InRelease
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Installing:
  rkhunter

Installing dependencies:
  bsd-mailx  exim4-base  exim4-config  exim4-daemon-light  liblockfile1  unhide  unhide.rb

Suggested packages:
```

Install chkrootkit

sudo apt install chkrootkit -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install chkrootkit -y

The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Installing:
  chkrootkit

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 34
  Download size: 321 kB
  Space needed: 989 kB / 52.6 GB available
```

Then Run Scan Directly

sudo rkhunter --check --skip-keypress

```
┌──(kali㉿kali)-[~]
└─$ sudo rkhunter --check --skip-keypress

[ Rootkit Hunter version 1.4.6 ]

Checking system commands ...

  Performing 'strings' command checks
    Checking 'strings' command                              [ OK ]

  Performing 'shared libraries' checks
    Checking for preloading variables                       [ None found ]
    Checking for preloaded libraries                        [ None found ]
    Checking LD_LIBRARY_PATH variable                       [ Not found ]

  Performing file properties checks
    Checking for prerequisites                              [ OK ]
    /usr/sbin/adduser                                       [ OK ]
    /usr/sbin/chroot                                        [ OK ]
    /usr/sbin/cron                                          [ OK ]
    /usr/sbin/depmod                                        [ OK ]
```

Run chkrootkit Scan

sudo chkrootkit

```
┌──(kali㉿kali)-[~]
└─$ sudo chkrootkit

ROOTDIR is `/'
Checking `amd' ...                                          not found
Checking `basename' ...                                     not infected
Checking `biff' ...                                         not found
Checking `chfn' ...                                         not infected
Checking `chsh' ...                                         not infected
Checking `cron' ...                                         not infected
Checking `crontab' ...                                      not infected
Checking `date' ...                                         not infected
Checking `du' ...                                           not infected
Checking `dirname' ...                                      not infected
Checking `echo' ...                                         not infected
Checking `egrep' ...                                        not infected
Checking `env' ...                                          not infected
Checking `find' ...                                         not infected
Checking `fingerd' ...                                      not found
Checking `gpm' ...                                          not found
Checking `grep' ...                                         not infected
Checking `hdparm' ...                                       not infected
```

Check for Suspicious Kernel Modules

lsmod

```
┌──(kali㉿kali)-[~]
└─$ lsmod

Module                  Size  Used by
mptcp_diag             16384  0
xsk_diag               12288  0
vsock_diag             12288  0
raw_diag               12288  0
unix_diag              12288  0
af_packet_diag         12288  0
netlink_diag           12288  0
tcp_diag               20480  0
udp_diag               12288  0
inet_diag              20480  4 tcp_diag,mptcp_diag,r
xt_recent              24576  0
```

Check Hidden Processes

sudo ps auxf

```
┌──(kali㉿kali)-[~]
└─$ sudo ps auxf

USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           2  0.0  0.0      0     0 ?        S    08:26   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    08:26   0:00  \_ [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-netns]
root          11  0.1  0.0      0     0 ?        I    08:26   0:02  \_ [kworker/0:1-ata_sff]
root          13  0.0  0.0      0     0 ?        I<   08:26   0:00  \_ [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    08:26   0:00  \_ [ksoftirqd/0]
root          15  0.1  0.0      0     0 ?        I    08:26   0:02  \_ [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    08:26   0:00  \_ [rcu_exp_par_gp_kthread_worker/0]
```

Compare with:

sudo top

```
┌──(kali㉿kali)-[~]
└─$ sudo top
top - 08:59:17 up 32 min,  1 user,  load average: 0.29, 0.67, 0.52
Tasks: 185 total,   1 running, 184 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.8 us,  2.3 sy,  0.0 ni, 94.0 id,  1.4 wa,  0.0 hi,  0.5 si,  0.0 st
MiB Mem :   1971.7 total,    110.6 free,   1062.6 used,    993.8 buff/cache
MiB Swap:    953.7 total,    727.3 free,    226.4 used.    909.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    813 root      20   0  417316 114656  54472 S   3.6   5.7   0:58.10 Xorg
   1521 kali      20   0  514964  37948  20988 S   1.0   1.9   0:01.06 xfce4-panel
   1488 kali      20   0  559392  85052  55144 S   0.7   4.2   0:12.24 xfwm4
   1897 kali      20   0  661640  46392  31968 S   0.7   2.3   0:06.39 qterminal
    588 root      20   0  353140   2676   2408 S   0.3   0.1   0:03.17 VBoxDRMClient
    592 root      20   0  292012   3224   2812 S   0.3   0.2   0:01.37 VBoxService
    857 mysql     20   0 1154424  48764   8848 S   0.3   2.4   0:02.59 mariadbd
   1411 kali      20   0  215100   3176   2964 S   0.3   0.2   0:02.06 VBoxClient
   1418 kali      20   0  215616   3024   2952 S   0.3   0.1   0:06.61 VBoxClient
   1430 kali      20   0  215288   3236   3136 S   0.3   0.2   0:00.70 VBoxClient
   1545 kali      20   0  296628  21412  13908 S   0.3   1.1   0:08.49 wrapper-2.0
```

Check Unexpected Open Ports

sudo ss -tulnp

```
┌──(kali㉿kali)-[~]
└─$ sudo ss -tulnp

Netid       State       Recv-Q      Send-Q            Local Address:Port              Peer Address:Port      Process
tcp         LISTEN      0           80                    127.0.0.1:3306                  0.0.0.0:*          users:(("mariadbd",pid=857,fd=27))
tcp         LISTEN      0           128                     0.0.0.0:22                    0.0.0.0:*          users:(("sshd",pid=9315,fd=6))
tcp         LISTEN      0           4096                  127.0.0.1:35403                 0.0.0.0:*          users:(("containerd",pid=772,fd=9))
tcp         LISTEN      0           128                        [::]:22                       [::]:*          users:(("sshd",pid=9315,fd=7))

┌──(kali㉿kali)-[~]
```

Verify System File Integrity (Optional)

sudo debsums -s

if not found

sudo apt update

sudo apt install debsums -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
sudo apt install debsums -y

Hit:1 http://http.kali.org/kali kali-rolling InRelease
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Installing:
  debsums

Installing dependencies:
  libfile-fnmatch-perl

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 34
  Download size: 51.5 kB
  Space needed: 154 kB / 52.6 GB available
```

Verify System File Integrity (Optional)

sudo debsums -s -e

```
┌──(kali㉿kali)-[~]
└─$ sudo debsums -s -e

debsums: changed file /etc/apache2/apache2.conf (from apache2 package)
debsums: changed file /etc/security/pwquality.conf (from libpwquality-common package)
debsums: missing file /etc/gvm/notus-scanner.toml (from notus-scanner package)
```

4. Generate a score/report based on CIS benchmarks.

The CIS (Center for Internet Security) Benchmarks are globally recognized security configuration guidelines designed to harden operating systems, software, and network devices. They provide detailed recommendations for securing systems against common threats. A CIS benchmark assessment evaluates system configurations such as password policies, firewall settings, file permissions, services, and logging configurations. After scanning, a score or report is generated that shows compliance percentage and highlights vulnerabilities or misconfigurations. Tools like Lynis, OpenSCAP, and CIS-CAT can be used to generate compliance reports aligned with CIS standards. Regular benchmarking improves system security posture and helps meet compliance requirements.

Install Lynis

sudo apt update

sudo apt install lynis -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
sudo apt install lynis -y

Hit:1 http://http.kali.org/kali kali-rolling InRelease
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Installing:
  lynis

Installing dependencies:
  menu

Suggested packages:
  apt-listbugs  debsecan  tripwire  samhain  aide  menu-l10n

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 34
  Download size: 631 kB
  Space needed: 3,293 kB / 52.6 GB available
```

Run System Audit

sudo lynis audit system

```
┌──(kali㉿kali)-[~]
└─$ sudo lynis audit system

[ Lynis 3.1.6 ]

################################################################################
  Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
  welcome to redistribute it under the terms of the GNU General Public License.
  See the LICENSE file for details about using this software.

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)
################################################################################


[+] Initializing program
------------------------------------
  - Detecting OS ...                                          [ DONE ]
  - Checking profiles ...                                     [ DONE ]
```

View CIS-Based Security Score

At the end of the scan, you will see:

Hardening index : XX [0-100]

View Detailed Report File

Lynis saves reports in:

sudo cat /var/log/lynis-report.dat

```
┌──(kali㉿kali)-[~]
└─$ sudo cat /var/log/lynis-report.dat

# Lynis Report
report_version_major=1
report_version_minor=0
report_datetime_start=2026-02-19 09:10:44
auditor=[Not Specified]
lynis_version=3.1.6
os=Linux
os_name=Kali Linux
os_fullname=Kali GNU/Linux Rolling
os_version=Rolling release
linux_version=Kali
os_kernel_version=6.18.9+kali
os_kernel_version_full=6.18.9+kali-amd64
hostname=kali
test_category=all
test_group=all
plugin_directory=/etc/lynis/plugins
lynis_update_available=0
```

Generate Report File for Submission

sudo lynis audit system > cis_report.txt

```
┌──(kali㉿kali)-[~]
└─$ sudo lynis audit system > cis_report.txt
```

Check report:

cat cis_report.txt

```
┌──(kali㉿kali)-[~]
└─$ cat cis_report.txt

[ Lynis 3.1.6 ]

################################################################################
  Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
  welcome to redistribute it under the terms of the GNU General Public License.
  See the LICENSE file for details about using this software.

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)
################################################################################


[+] Initializing program
------------------------------------
  - Detecting OS ...                                          [ DONE ]
  - Checking profiles ...                                     [ DONE ]

  ---------------------------------------------------------
  Program version:           3.1.6
  Operating system:          Linux
  Operating system name:     Kali Linux
  Operating system version:  Rolling release
  End-of-life:               UNKNOWN
  Kernel version:            6.18.9+kali
  Hardware platform:         x86_64
  Hostname:                  kali
  ---------------------------------------------------------
  Profiles:                  /etc/lynis/default.prf
```

5. Recommend hardening actions.

System hardening is the process of securing a computer system by reducing its attack surface and eliminating vulnerabilities. It involves applying security best practices such as disabling unnecessary services, configuring firewalls, enforcing strong password policies, securing SSH access, updating software, and setting proper file permissions. Hardening ensures that only essential services are running and that the system follows security standards like CIS benchmarks. Regular system updates, monitoring logs, restricting user privileges, and enabling auditing mechanisms significantly improve the security posture of a Linux system and protect it against cyber threats.

Update System Packages

sudo apt update

sudo apt upgrade -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
sudo apt upgrade -y

Hit:1 http://http.kali.org/kali kali-rolling InRelease
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Upgrading:
  gparted         libhdf5-310     libnetfilter-conntrack3  manpages-dev        mingw-w64-x86-64-dev  python3-celery
  gparted-common  libhdf5-hl-310  libnvme1t64              mingw-w64-common    mupdf-tools           python3-scipy
  libgc1          libmupdf27.0    manpages                 mingw-w64-i686-dev  python3-arc4

Not upgrading:
  colord    libcolorhug2  libgbm1         libglx-mesa0  libgtk-4-bin    mesa-libgallium  python-tables-data  python3-tables      yelp
  graphviz  libegl-mesa0  libgl1-mesa-dri  libgtk-4-1    libgtkmm-4.0-0  nodejs           python3-pygraphviz  python3-tables-lib
```

Enable and Configure Firewall (UFW)

sudo apt install ufw -y

sudo ufw default deny incoming

sudo ufw default allow outgoing

sudo ufw allow ssh

sudo ufw enable

sudo ufw status verbose

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install ufw -y
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow ssh
sudo ufw enable
sudo ufw status verbose

ufw is already the newest version (0.36.2-9).
The following packages were automatically installed and are no longer re
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
Skipping adding existing rule
Skipping adding existing rule (v6)
Firewall is active and enabled on system startup
Status: active
```

Disable Unnecessary Services

List running services:

sudo systemctl list-units --type=service --state=running

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl list-units --type=service --state=running

  UNIT                          LOAD   ACTIVE SUB     DESCRIPTION
  accounts-daemon.service       loaded active running Accounts Service
  colord.service                loaded active running Manage, Install and Generate Color Profiles
  containerd.service            loaded active running containerd container runtime
  cron.service                  loaded active running Regular background program processing daemon
  dbus.service                  loaded active running D-Bus System Message Bus
  docker.service                loaded active running Docker Application Container Engine
  fail2ban.service              loaded active running Fail2Ban Service
  firewalld.service             loaded active running firewalld - dynamic firewall daemon
  getty@tty1.service            loaded active running Getty on tty1
  haveged.service               loaded active running Entropy Daemon based on the HAVEGE algorithm
  lightdm.service               loaded active running Light Display Manager
  mariadb.service               loaded active running MariaDB 11.8.5 database server
  ModemManager.service          loaded active running Modem Manager
  NetworkManager.service        loaded active running Network Manager
```

Disable unwanted service (example):

sudo systemctl stop apache2

sudo systemctl disable apache2

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl stop apache2
sudo systemctl disable apache2

Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable apache2
```

Secure SSH Configuration

Edit SSH config:

sudo nano /etc/ssh/sshd_config

Change/add:

PermitRootLogin no

PasswordAuthentication no

MaxAuthTries 3

```
  GNU nano 8.7                                                    /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
```

Restart SSH:

sudo systemctl restart ssh

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart ssh
```

Enforce Strong Password Policy

Install password quality module:

sudo apt install libpam-pwquality -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install libpam-pwquality -y

The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Installing:
  libpam-pwquality

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 17
  Download size: 13.3 kB
  Space needed: 42.0 kB / 52.6 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libpam-pwquality amd64 1.4.5-5+b1 [13.3 kB]
Fetched 13.3 kB in 1s (21.7 kB/s)
```

Edit:
sudo nano /etc/security/pwquality.conf
Example settings:
minlen = 12
ucredit = -1
lcredit = -1
dcredit = -1
ocredit = -1

Set Correct File Permissions
sudo chmod 644 /etc/passwd
sudo chmod 640 /etc/shadow

```
┌──(kali㉿kali)-[~]
└─$ sudo chmod 644 /etc/passwd
sudo chmod 640 /etc/shadow

┌──(kali㉿kali)-[~]
```

Enable Automatic Security Updates
sudo apt install unattended-upgrades -y

sudo dpkg-reconfigure unattended-upgrades

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install unattended-upgrades -y
sudo dpkg-reconfigure unattended-upgrades

unattended-upgrades is already the newest version (2.12+nmu1).
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
```

Install Intrusion Detection Tool

sudo apt install rkhunter -y

sudo rkhunter --check --skip-keypress

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install rkhunter -y
sudo rkhunter --check --skip-keypress

rkhunter is already the newest version (1.4.6-13).
The following packages were automatically installed and are no longer required:
  librubberband2  libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
```

Check Open Ports

sudo ss -tulnp

```
┌──(kali㉿kali)-[~]
└─$ sudo ss -tulnp

Netid       State       Recv-Q      Send-Q            Local Address:Port              Peer Address:Port      Process
tcp         LISTEN      0           80                    127.0.0.1:3306                  0.0.0.0:*          users:(("mariadbd",pid=857,fd=27))
tcp         LISTEN      0           128                     0.0.0.0:22                    0.0.0.0:*          users:(("sshd",pid=227205,fd=6))
tcp         LISTEN      0           4096                  127.0.0.1:35403                 0.0.0.0:*          users:(("containerd",pid=772,fd=9))
tcp         LISTEN      0           128                        [::]:22                       [::]:*          users:(("sshd",pid=227205,fd=7))
```

6. Conclusion :-

The performed security tasks—including firewall configuration, service management, SSH hardening, verification of critical file permissions, rootkit detection, integrity checking, and CIS benchmark assessment—collectively strengthen the overall security posture of the Linux system. By minimizing unnecessary services, restricting

network access, enforcing secure authentication policies, and regularly auditing system configurations, the attack surface is significantly reduced. Tools such as Lynis, rkhunter, and debsums help identify vulnerabilities and ensure system integrity. Implementing these hardening measures not only protects against unauthorized access and malware but also aligns the system with industry-recognized security best practices like CIS benchmarks. Continuous monitoring and periodic security assessments are essential to maintain a secure and resilient computing environment.